

Ultimate HR Attrition Project Viva Handbook

Project Objective

Topic	Explanation
Goal	Predict employee attrition
Target	Attrition
Classes	0 = Stay, 1 = Leave
Problem Type	Binary Classification
Learning Type	Supervised Learning

Libraries

Library	Purpose
pandas	Table handling
numpy	Math operations
matplotlib	Graph plotting
seaborn	Visualization
sklearn	Machine learning
xgboost	Boosting algorithm

Preprocessing

Step	Why Used	Alternative
Duplicate Removal	Avoid biased learning	Keep duplicates
Missing Values	Models cannot process NaN	Delete rows
Label Encoding	Convert text to numbers	One Hot Encoding
Drop ID Columns	No predictive value	Keep columns
Scaling	Equal feature magnitudes	MinMaxScaler

Train Test Split

Parameter	Meaning
test_size=0.2	20 percent testing
random_state=42	Reproducibility

Models Used

Model	Why Used	Strength
Logistic Regression	Baseline model	Simple
Decision Tree	Rule-based learning	Interpretable
Random Forest	Ensemble learning	Stable
KNN	Similarity learning	Simple
XGBoost	Boosting	High performance

Hyperparameters

Hyperparameter	Meaning
max_iter=1000	Optimization iterations
criterion='gini'	Split impurity
max_depth=5	Prevent overfitting
n_estimators=100	Number of trees
n_neighbors=5	Nearest neighbors
eval_metric='logloss'	Probability error

Evaluation Metrics

Metric	Meaning
Accuracy	Overall correctness
Precision	Correct positive predictions
Recall	Detect actual attrition
F1 Score	Precision recall balance
ROC-AUC	Classifier quality

Confusion Matrix

Term	Meaning
TP	Correct attrition prediction
TN	Correct stay prediction
FP	False alarm
FN	Missed attrition

SMOTE

Aspect	Explanation
Purpose	Balance minority class
Problem	Class imbalance
Benefit	Improved recall
Alternative	Oversampling or undersampling

Feature Importance

Feature	Reason
MonthlyIncome	Salary affects retention
OverTime	Burnout increases attrition
Age	Age influences job switching
DistanceFromHome	Travel stress

Project Flow

Step	Purpose
Load Data	Bring dataset into memory
Clean Data	Improve quality
Encode Data	Convert text to numbers
Scale Data	Normalize features
Train Models	Learn patterns
Evaluate	Measure performance
SMOTE	Balance classes